

Basic Statistical Formulae

Major Formulae

The mean is the sum of the scores divided by the number of scores:

$$\bar{X} = \frac{\sum X}{N}$$

Where N is the sample size.

The variance is the sum of the squared deviations of the scores from the mean, divided by the number of scores:

$$\frac{\sum (X - \bar{X})^2}{N}$$

This “quick” version is mathematically equivalent, but faster to compute by hand

$$\frac{\sum X^2 - \frac{(\sum X)^2}{N}}{N}$$

The standard deviation is the square root of the variance:

$$SD = \sqrt{\frac{\sum (X - \bar{X})^2}{N}} = \sqrt{\frac{\sum X^2 - \frac{(\sum X)^2}{N}}{N}}$$

Note, when computing the standard deviation (or variance) of a sample of data (i.e., not the entire population) the denominator used is N-1:

$$SD = \sqrt{\frac{\sum (X - \bar{X})^2}{N - 1}} = \sqrt{\frac{\sum X^2 - \frac{(\sum X)^2}{N}}{N - 1}}$$